

BT-302 SUBJECTIVE MID TERM-

100% GUESS PAPER 2024

IMMUNOLOGY BY

SULMAN ALI GUJJAR (ZOOLOGIST)

0341-7547044

SUBJECTIVES

What are monoclonal antibodies?

The antibodies which are produced from a single antibody producing B-cell are called as monoclonal antibodies. These antibodies also able to bind with single and unique antigen binding site (epitope). Monoclonal antibodies mainly consist of single subtype of IgG for example IgG1, IgG2 & IgG3.

Define phagocytosis. Explain the process of phagocytosis:

Phagocytosis is a vital cellular process where specialized cells called phagocytes engulf and digest solid particles like bacteria and cellular debris. The process involves several stages:

1. **Chemotaxis:** Phagocytes move towards infection sites due to chemical signals.
2. **Attachment:** Phagocytes adhere to foreign particles using specific receptors.
3. **Engulfment:** Phagocytes extend their membrane to surround the particle, forming a phagosome.
4. **Phagosome Formation:** The phagosome matures and fuses with lysosomes.
5. **Digestion:** Lysosomal enzymes break down the phagosome's contents.
6. **Exocytosis:** Indigestible material is expelled from the cell.

Differentiate between T-dependent and T-independent antigens.

T-cell dependent antigens: These are antigens which need the help of T-cells for mounting an immune response otherwise these cannot. T-dependent requires the help of T-cells for activation of B-cells. Such kinds of antigens are protein in nature. Moreover, these antigens carry variety of antigenic determinants (epitopes) with few copies.

T-Cell independent antigens: These antigens are those antigens which can directly activate B-cells without the help of T-cells. Such kinds of antigens are non-protein and usually are polysaccharides in nature. These antigens carry same kind of epitopes in polymeric form (Figure 3.3). Moreover, such kinds of antigens are most resistant to degradation.

Activation of cytotoxic t cells:

Cytotoxic T cells, or CD8+ T cells, become activated when their receptors recognize antigens on infected cells presented by MHC-I molecules. This recognition triggers a signaling cascade, leading to T cell activation, proliferation, and differentiation into killer cells. These activated cytotoxic T cells then target and eliminate infected or abnormal cells through various mechanisms, ultimately helping to control infections and prevent the spread of pathogens.

Difference between T-CELLS and B-CELLS:

1. T Cells: These cells are central to the adaptive immune system and play various roles, such as activating other immune cells and directly attacking infected or abnormal cells.
2. B Cells: They are responsible for producing antibodies, which are proteins that help identify and neutralize pathogens.

Three components of coagulation:

Coagulation, also known as clotting, is the process by which blood changes from a liquid to a gel, forming a blood clot to stop bleeding after an injury.

1= Platelets

These cellular blood components form a plug at the injury site, initiating clot formation (primary hemostasis).

2= Coagulation Factors

These proteinaceous components, including fibrinogen and prothrombin, participate in a cascade to form fibrin strands that strengthen the platelet plug.

3= Fibrin

This protein forms threads that cross-link and stabilize the platelet plug, leading to clot formation (secondary hemostasis).

Define heptane:

Heptane: Heptane is a hydrocarbon compound with the chemical formula C_7H_{16} . It is a colorless, flammable liquid that is commonly used as a solvent in laboratories and industrial processes.

Lymphocytes recirculation?

Lymphocyte recirculation is the process by which lymphocytes, including T cells and B cells, constantly migrate between the blood and lymphoid tissues. This movement facilitates immune surveillance and responses throughout the body. Lymphocytes exit the bloodstream into lymphoid organs like lymph nodes, spleen, and Peyer's patches via high endothelial venules. Within these tissues, they encounter antigens, undergo activation, and differentiate into effector cells. After completing their immune functions, lymphocytes re-enter circulation through lymphatic vessels, allowing them to migrate to other sites to continue immune surveillance and responses.

What is affinity maturation?

The affinity of antibodies depend upon the dosage of antigen, the low dosage of antigen causes increase in the affinity of antibodies which is called as affinity maturation.

What are the characteristics of antibodies?

Antibodies have several characteristics:

1. **Antigen Binding Properties:** Present on B-cell membranes and secreted by plasma cells, conferring antigenic specificity and leading to B-cell proliferation.
2. **Effectors of Humoral Immunity:** Circulate in the blood, neutralizing or marking antigens for elimination.

3. **Structural Features and Effector Functions:** Share structural features, bind to antigens, and participate in immune response.
4. **Heterogeneity:** Produced in response to multiple epitopes on complex antigens

If a newly virus or bacteria enter into the grown man body then which of the immune system will overcome the disease? Explain briefly

The adaptive immune system, particularly involving B cells and T cells, will overcome the disease. B cells produce antibodies targeting the pathogen, while T cells destroy infected cells. This response is specific to the new virus or bacteria, ensuring effective elimination and long-term immunity.

On which basis the antibodies are divided into types ? Explain the types and subtypes of the antibodies.

Antibodies are divided into types based on their heavy chain structure and function. The five main types are:

1. **IgG:** Monomer, provides long-term immunity and crosses the placenta. Subtypes: IgG1, IgG2, IgG3, IgG4.
2. **IgA:** Usually a dimer, protects mucosal surfaces and is found in secretions. Subtypes: IgA1, IgA2.
3. **IgM:** Pentamer, first antibody produced in the initial immune response, no subtypes.
4. **IgE:** Monomer, involved in allergic reactions and defense against parasites, no subtypes.
5. **IgD:** Monomer, acts as a receptor on naïve B cells, no subtypes.

What is the antibodies processing and presentation?

Antigen Processing and Presentation

Antigen processing and presentation are crucial steps in the immune response, involving the association of protein fragments with major histocompatibility complex (MHC) molecules for T cell recognition.

Process Overview

- **Proteolysis:** Proteins are fragmented within cells.
- **MHC Association:** Fragments associate with MHC molecules.
- **Expression:** Peptide-MHC complexes are displayed on the surface of antigen-presenting cells (APCs) like macrophages.

Class I MHC

- **Source:** Presents degradation products from intracellular (endogenous) proteins in the cytosol.
- **Recognition:** Recognized by cytotoxic T cells (CD8+), leading to cell killing.

Class II MHC

- **Source:** Presents fragments from extracellular (exogenous) proteins in intracellular compartments.
- **Recognition:** Recognized by helper T cells (CD4+), leading to immune activation and coordination.

Differentiate between b cell receptors and t cell receptor:

B cell receptors (BCRs) are on B cells, recognize intact antigens, and initiate humoral immunity by producing antibodies. T cell receptors (TCRs) are on T cells, recognize peptide fragments

presented by MHC molecules, and are crucial for cell-mediated immunity, activating T cells and regulating immune responses.

Cell of cellular barrier:

Cells of the cellular barrier are key components of the immune system's first line of defense.

1. **Neutrophils:** Rapidly respond to infections, phagocytose bacteria, release antimicrobials.
2. **Macrophages:** Phagocytose pathogens, present antigens, secrete cytokines.
3. **Dendritic Cells:** Capture and present antigens to T cells.
4. **NK Cells:** Destroy infected/cancerous cells by inducing apoptosis.
5. **Eosinophils:** Combat parasitic infections, participate in allergic reactions.

Briefly describes the sub region of the variable region in the antibody

The variable region of antibodies, present on both heavy and light chains, has two sub-regions:

1. **Hypervariable (HVR):** Characterized by variability in amino acid sequences, forming the complementarity determining region (CDR).
2. **Framework Regions:** Located between hypervariable regions, they group and subgroup immunoglobulins and are products of different variable region genes.

Effector function of complement proteins:

The secondary effector functions of complement proteins enhance immune responses:

1. **Opsonization:** Proteins like C3b, iC3b, and C4b tag microbes, enhancing phagocytosis by forming complexes recognized by complement receptors on phagocytes.
2. **Chemotaxis:** Proteins like C5a attract phagocytic cells to infection sites, activating neutrophils, basophils, and macrophages, and inducing adhesion molecule expression on endothelial cells for phagocyte migration.
3. **Anaphylaxis:** Proteins like C4a, C3a, and C5a trigger exaggerated immune responses by causing basophils and mast cells to degranulate, releasing mediators that lead to smooth muscle contraction, vasodilation, and bronchoconstriction.

Factors that affect antibody and antigen test:

factors affecting antigen and antibody tests include:

1. **Affinity:** The strength of binding between a single antigenic determinant and a single antibody site. Higher affinity results in more stable interactions.
2. **Avidity:** The overall strength of binding between multivalent antigens and antibodies, influenced by the valency of both. Greater avidity leads to more stable and easily detectable reactions.
3. **Antigen & Antibody Ratio:** The ratio determines the nature of the complex, with equivalence facilitating lattice formation and larger complexes being more detectable.
4. **Physical Form:** The physical form of the antigen affects the reaction type, with particulate antigens causing agglutination and soluble antigens causing precipitation.

What is secondary lymphoid organs.

Secondary lymphoid organs, like lymph nodes, spleen, and mucosal-associated lymphoid tissues (MALT), including gut-associated lymphoid tissue (GALT), trap antigens and facilitate interaction between mature lymphocytes and antigens. They play a crucial role in the adaptive immune response by facilitating immune cell activation and proliferation.

ELISA Advantages and differentiate between direct and indirect ELISA.

ELISA Applications

ELISA, a solid-phase immunoassay, is versatile in various applications:

1. **Viral Antigen Detection:** Identifying viral antigens like Hepatitis B surface Antigen (HBsAG) in patient serum.
2. **Antibody Determination:** Detecting antibodies against viruses such as anti-HCV antibodies.
3. **Serum Cytokine Levels:** Measuring levels of serum cytokines like IL-1 and TNF- α .

Direct ELISA

Principle: Detects unknown antibodies.

1. **Coating:** Known antigens coat the solid phase.
2. **Reaction:** Patient serum with unknown antibodies reacts.
3. **Detection:** Enzyme-coated secondary antibody reacts, indicating antibody presence.

Indirect ELISA

Principle: Determines unknown antigens.

1. **Coating:** Known antibodies coat the solid phase.
2. **Reaction:** Patient serum with unknown antigens reacts.
3. **Sandwich Formation:** Enzyme-coated secondary antibody binds, indicating antigen presence.

Function of granulocytes and Megakaryocytes:

Megakaryocytes are large bone marrow cells responsible for platelet production. They serve as the source of platelets, which are vital for blood clotting (hemostasis), wound healing, and modulating immune responses by interacting with other immune cells.

Granulocytes are a type of white blood cell with granules in their cytoplasm. They include:

1. **Neutrophils:** First responders to infection, engulf and destroy bacteria and fungi, crucial for innate immunity.
2. **Basophils:** Involved in allergic responses, release histamine contributing to inflammation and allergy symptoms.
3. **Mast Cells:** Release histamine during inflammation and allergic reactions, important for defense against pathogens and allergic responses.

T-cell receptor:

T-Cell Receptors (TCR) are essential for antigen recognition by T cells. They primarily recognize protein antigens presented by major histocompatibility complex (MHC) molecules on nucleated cells. TCRs categorize T cells based on the MHC molecules they recognize: helper T cells bind to peptides presented by class II MHC molecules, while cytotoxic T cells bind to peptides presented by class I MHC molecules. Antigen recognition by TCRs requires presentation by MHC molecules, and their specificity is dependent on the class of MHC molecules presenting the peptide fragments.

Short difference between TCR and bcr:

T-Cell Receptors (TCR) primarily recognize protein antigens presented by MHC molecules on nucleated cells, categorizing T cells based on MHC recognition. In contrast, **B-Cell Receptors (BCR)** recognize soluble antigens directly, facilitating antibody production. TCRs rely on MHC presentation, while BCRs reflect the specificity of the antibodies they produce.

What is the role of immune system in intracellular pathogens:

The immune system employs distinct strategies to combat intracellular pathogens. Granulocytes, including neutrophils, basophils, and mast cells, act in the innate immune response. Neutrophils rapidly respond to infections, while basophils and mast cells release histamine during allergic reactions. Megakaryocytes produce platelets essential for blood clotting and wound healing. In combating intracellular pathogens, cell-mediated immunity led by T lymphocytes is crucial. Cytotoxic T lymphocytes recognize antigens presented by infected cells' class I MHC molecules, controlling cytosolic pathogens like viruses, while helper T lymphocytes recognize antigens from vesicular pathogens such as intracellular bacteria.

Name the cells which are involved in immune system:

1= T Cells, 2= B Cells, 3= Natural Killer (NK) Cells, 4= Macrophages, 5= Dendritic Cells, 6= Neutrophils, 7= Eosinophils, 8= Basophils and Mast Cells

Briefly describe the monoclonal antibodies usage:

Monoclonal antibodies are utilized in disease management, immune cell identification, tumor typing and analysis, cancer treatment, and cell sorting. They aid in preventing, diagnosing, and treating diseases, targeting cell surface markers, classifying tumors, studying solid tumors, and separating cells using techniques like Fluorescence Activated Cell Sorting (FACS).

Explain basic structure of antibodies:

Antibodies consist of two light (L) chains and two heavy (H) chains forming a heterodimer structure. Each chain has variable (V) and constant (C) regions. Variable regions determine antigen specificity and contain complementarity-determining regions (CDRs) for antigen binding. Constant regions maintain sequence consistency within antibody classes and facilitate carbohydrate attachment. A hinge region provides flexibility, allowing the antibody to change shape upon antigen binding. Heavy chains typically have four domains, while light chains have two, each with an intra-chain disulfide bond. Oligosaccharides attached to the CH₂ domain enhance stability and modulate immune responses.

Describe the types of two adjuvants

Adjuvants, enhancing immune responses to weak antigens, exist in two forms:

1. **Chemical Adjuvants:** Aluminum salts (Alum) like those in DTP vaccines. Alum functions by gradually releasing the antigen, increasing interaction with Toll-like receptors (TLRs) and activating mononuclear phagocytes. This prompts heightened cytokine secretion, bolstering the immune response.
2. **Biological Adjuvants:** Derived from various bacteria, these substances stimulate the immune system through bacterial components. They augment the immune response to vaccine antigens by leveraging their bacterial properties.

Oxygen dependent & independent intracellular killing:

Intracellular killing of bacteria occurs through two main methods: oxygen-independent, where phagocytes release bactericidal enzymes, and oxygen-dependent, involving a respiratory burst. In the latter, increased glucose and oxygen consumption after phagocytosis leads to the production of oxygen-containing bactericidal radicals, including halide ions, which kill bacteria.

Polyclonal Antibodies:

Polyclonal antibodies are sourced from multiple antibody-producing B lymphocytes. They exhibit specificity to multiple epitopes on the same antigen and are obtained from serum containing antibodies with diverse affinities. Mainly belonging to the IgG class, polyclonal antibodies are valuable tools in research and diagnostics due to their ability to recognize various antigenic sites, enhancing the detection and quantification of target molecules in biological samples.

Difference between exogenous and endogenous antigen processing

Endogenous antigens originate within cells, such as viruses, processed via proteasomes, and presented on MHC I molecules on all nucleated cells. Exogenous antigens, like bacteria, are internalized through endocytosis, processed by proteases in endosomes, and presented on MHC II molecules mainly by antigen-presenting cells like macrophages, dendritic cells, and B-cells.

Define antigen, antibodies and heptane ?

1= Antigen A substance that reacts with the products of specific immune response.

2= Immunogen A substance that induces specific immune response.

3= Hapten A substance that is nonimmunogenic but can react with products of specific immune response

Clonal selection of lymphocytes and write it's function?

Definition: Clonal selection is the process by which lymphocytes with unique receptor specificity undergo activation, differentiation, and selection against non-self antigens.

Function: It ensures the generation of a diverse lymphocyte repertoire capable of recognizing and responding to foreign molecules while maintaining tolerance to self-antigens, thereby facilitating effective immune responses and preventing autoimmune reactions.

Brief note on hybridoma.

Hybridomas are created by fusing a lymphoid tumor cell with a normal B-lymphocyte. This hybrid cell line combines the immortality of the tumor cell with the single-specificity antibody production capability of the B-lymphocyte, enabling continuous production of monoclonal antibodies.

How should be chemical nature of good immunogen?

A good immunogen should be complex and large, typically with a molecular weight greater than 10,000 daltons, to ensure it can be effectively recognized and processed by the immune system. It should be foreign to the host and chemically diverse, with multiple epitopes to elicit a strong immune response. Immunogens must also be stable enough to be recognized but degradable by antigen-presenting cells for presentation on MHC molecules.

Anatomical Barriers of immune system:

Anatomical barriers include the skin and squamous epithelial membranes, which provide physical protection. Mechanical factors such as desquamation, peristalsis, ciliary movement, and the flushing actions of tears and saliva remove microbes. Mucus in the respiratory and gastrointestinal tracts traps pathogens. Chemical factors include fatty acids in the skin, lysozymes, phospholipases, antimicrobial peptides (defensins), low pH in sweat and GIT secretions, and surfactants in the lungs. Normal flora in the skin and GIT prevent bacterial colonization by secreting toxins and competing for nutrients.

How the fragments of antibodies are obtained??

Antibody fragments are obtained through enzyme digestion. **Papain digestion** of immunoglobulin G (IgG) produces two **Fab fragments** (each 45,000 molecular weight) that bind antigens and one **Fc fragment** (50,000 molecular weight) that does not bind antigens but crystallizes during cold storage. **Pepsin digestion** further separates the antigen-binding properties from the rest of the antibody, isolating the parts responsible for binding antigens from other segments. These processes allow the study and utilization of specific antibody functions.

Neutralization against extracellular pathogens:

Neutralization involves antibodies binding to surface antigens on pathogens, blocking their interaction with host cells. This prevents pathogens from attaching and entering host cells, neutralizes toxins they produce, and facilitates their clearance by phagocytes.

Cells produce by myeloid Lineage:

Myeloid stem cells give rise to various types of blood cells, including:

1. Erythrocytes (Red Blood Cells) = Function: Transport oxygen and carbon dioxide.
2. Megakaryocytes (Platelets)= Function: Blood clotting and wound repair.
3. Granulocytes: =
 - Neutrophils: Phagocytosis and killing of bacteria.
 - Eosinophils: Combat parasitic infections and participate in allergic reactions.
 - Basophils: Release histamine and other mediators of inflammation.
4. Monocytes: Function: Differentiate into macrophages and dendritic cells, which are involved in phagocytosis and antigen presentation.

Define Phylaxis:

Phylaxis refers to the protection against infection and disease. It is a general term used to describe the body's defensive mechanisms, including the immune system's response to prevent and combat pathogens.

Chemical secretions of immunoglobulin:

Immunogens, which induce immune responses, are primarily proteins or large polysaccharides. They can also include lipoproteins, lipopolysaccharides, or nucleoproteins. Proteins are strong antigens due to their complex structure. Carbohydrates, nucleic acids, polypeptides, hormones, and lipids vary in immunogenicity.

Structure of TCR:

The T-cell receptor (TCR) is a surface receptor found on T-lymphocytes. It binds to antigens presented by major histocompatibility complex (MHC) molecules. Structurally, TCR resembles immunoglobulin and is part of the immunoglobulin superfamily. TCR is a heterodimer composed of one α and one β chain.

CFT:

CFT stands for Complement Fixation Test, a laboratory technique used to detect the presence of specific antibodies or antigens in a patient's blood serum. It measures the ability of antibodies to fix complement, which is a part of the immune system's response to pathogens.

What is MHC, difference between MHC-I and MHC-II?

Major Histocompatibility Complex (MHC) is a group of genes encoding cell surface proteins crucial for immune function. MHC class I molecules are present on all nucleated cells, presenting endogenous antigens to cytotoxic T cells, identifying infected or aberrant cells. MHC class II molecules, found mainly on antigen-presenting cells, present exogenous antigens to helper T cells, initiating immune responses against extracellular pathogens. Thus, MHC-I primarily interacts with cytotoxic T cells, while MHC-II interacts with helper T cells, reflecting their roles in cellular and humoral immunity, respectively.

Roll of somatic hyper mutation in diversity of antibodies

Somatic hypermutation is pivotal for antibody diversity, enabling the creation of antibodies with varied specificities against different pathogens. This process accelerates following exposure to infections, further enhancing antibody diversity. It occurs rapidly, approximately a million times faster than usual mutations. The protein AID, particularly, aids in altering the DNA of antibody genes, facilitating this diversification without impacting vital genetic components. Scientists continue to explore the intricacies of this mutation process for a comprehensive understanding.

Role of adjuvants in polyclonal antibodies production:

Adjuvants are crucial in polyclonal antibody production as they enhance the body's immune response to antigens. Common adjuvants like Freund's Complete and Incomplete Adjuvants stimulate immune responses, with the former containing heat-killed *Mycobacterium tuberculosis*.

for a robust reaction and the latter serving as booster doses without this component. However, adjuvants must be chosen carefully to balance immune stimulation and potential toxicity, ensuring optimal antibody production without adverse effects.

If new born baby acquired infection which type of immune system help him to overcome the infection? Briefly explain.

In newborns, the innate immune system plays a crucial role in combating infections. This system provides immediate, non-specific defense mechanisms against pathogens. Components such as skin barriers, mucous membranes, phagocytes, and natural killer cells act as the first line of defense, helping to prevent the entry and spread of pathogens.

Neutralization against extracellular pathogens:

Neutralization involves antibodies binding to surface antigens on pathogens, blocking their interaction with host cells. This prevents pathogens from attaching and entering host cells, neutralizes toxins they produce, and facilitates their clearance by phagocytes.

Cells produce by myeloid Lineage:

Myeloid stem cells give rise to various types of blood cells, including:

1. Erythrocytes (Red Blood Cells) = Function: Transport oxygen and carbon dioxide.
2. Megakaryocytes (Platelets) = Function: Blood clotting and wound repair.
3. Granulocytes: =
 - Neutrophils: Phagocytosis and killing of bacteria.
 - Eosinophils: Combat parasitic infections and participate in allergic reactions.
 - Basophils: Release histamine and other mediators of inflammation.
4. Monocytes: Function: Differentiate into macrophages and dendritic cells, which are involved in phagocytosis and antigen presentation.

Define Phylaxis:

Phylaxis refers to the protection against infection and disease. It is a general term used to describe the body's defensive mechanisms, including the immune system's response to prevent and combat pathogens.

Chemical secretions of immunoglobulin:

Immunogens, which induce immune responses, are primarily proteins or large polysaccharides. They can also include lipoproteins, lipopolysaccharides, or nucleoproteins. Proteins are strong antigens due to their complex structure. Carbohydrates, nucleic acids, polypeptides, hormones, and lipids vary in immunogenicity.

Structure of TCR:

The T-cell receptor (TCR) is a surface receptor found on T-lymphocytes. It binds to antigens presented by major histocompatibility complex (MHC) molecules. Structurally, TCR resembles immunoglobulin and is part of the immunoglobulin superfamily. TCR is a heterodimer composed of one α and one β chain.

CFT:

CFT stands for Complement Fixation Test, a laboratory technique used to detect the presence of specific antibodies or antigens in a patient's blood serum. It measures the ability of antibodies to fix complement, which is a part of the immune system's response to pathogens.

What is MHC, difference between MHC-I and MHC-II?

Major Histocompatibility Complex (MHC) is a group of genes encoding cell surface proteins crucial for immune function. MHC class I molecules are present on all nucleated cells, presenting endogenous antigens to cytotoxic T cells, identifying infected or aberrant cells. MHC

class II molecules, found mainly on antigen-presenting cells, present exogenous antigens to helper T cells, initiating immune responses against extracellular pathogens. Thus, MHC-I primarily interacts with cytotoxic T cells, while MHC-II interacts with helper T cells, reflecting their roles in cellular and humoral immunity, respectively.

Roll of somatic hyper mutation in diversity of antibodies

Somatic hypermutation is pivotal for antibody diversity, enabling the creation of antibodies with varied specificities against different pathogens. This process accelerates following exposure to infections, further enhancing antibody diversity. It occurs rapidly, approximately a million times faster than usual mutations. The protein AID, particularly, aids in altering the DNA of antibody genes, facilitating this diversification without impacting vital genetic components. Scientists continue to explore the intricacies of this mutation process for a comprehensive understanding.

Role of adjuvants in polyclonal antibodies production:

Adjuvants are crucial in polyclonal antibody production as they enhance the body's immune response to antigens. Common adjuvants like Freund's Complete and Incomplete Adjuvants stimulate immune responses, with the former containing heat-killed Mycobacterium tuberculosis for a robust reaction and the latter serving as booster doses without this component. However, adjuvants must be chosen carefully to balance immune stimulation and potential toxicity, ensuring optimal antibody production without adverse effects.

If new born baby acquired infection which type of immune system help him to overcome the infection? Briefly explain.

In newborns, the innate immune system plays a crucial role in combating infections. This system provides immediate, non-specific defense mechanisms against pathogens. Components such as skin barriers, mucous membranes, phagocytes, and natural killer cells act as the first line of defense, helping to prevent the entry and spread of pathogens. Long

VJD Recombination:

V(D)J recombination is a vital process in generating antibody diversity. It involves the joining of various gene segments, such as V and J in light chain genes and V, D, and J in heavy chain genes. Recombination is regulated by Recombination Signal Sequences (RSS), consisting of conserved heptamer and nonamer sequences separated by 2 and 1 turn signals. Enzymes like RAG1 and RAG2 facilitate this recombination process, crucial for the immune system's functionality.

Explain briefly pathogen associated molecular patterns.

The antigenic determinant recognized by innate immune system is purely non-specific in nature due to nonspecific attribute of innate immune system. These determinants are present on the surface of the pathogens in a form of various molecular configurations called as pathogen associated molecular patterns (PAMPs). PAMPs include cell membrane proteins, lipopolysaccharide and flagellular proteins. For these antigenic determinants immune cells carry pattern recognition receptors (PRRs) which bind non-specifically with corresponding PAMPs in order to generate immune response.